

Potential for small hydropower development in the Tano River Basin of Ghana using SWAT and RETScreen

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ARTICLE INFO

Keywords:

Small hydropower
Energy
SWAT
RETScreen
Run-of-River
Tano River Basin

ABSTRACT

Small hydropower (SHP) development offers promising sustainable energy solutions, particularly in regions with underutilised water resources. This study evaluates the potential for SHP development in the Tano River Basin (TRB) in Ghana using the Soil and Water Assessment Tool (SWAT) and RETScreen. SWAT was used to generate and simulate basin parameters for hydropower analysis, while the RETScreen tool was used for techno-economic and emissions analysis. The power and energy potential for the candidate sites were evaluated using flow and power duration curves. The results demonstrate that the TRB has potential for small hydropower development, with twenty-four sites along the Tano River having a firm energy potential ranging from 4.6 to 46.4 GWh. A detailed analysis of all the twenty-four sites revealed that HP9, HP11 and HP14 were the most promising, with Net Present Values (NPVs) of \$178,496,199, \$177,857,919 and \$175,248,329, respectively. An emissions analysis of these promising potential sites highlighted annual emissions savings averaging 66,382 tons of CO₂, equivalent to >140,000 barrels of oil not consumed. These findings underscore the potential for environmentally beneficial and economically viable SHP development, offering valuable insights for similar regional projects and contributing to Ghana's renewable energy and rural electrification efforts.

Introduction

Energy services are integral in meeting human needs (Brand-Correa & Steinberger, 2017). This is evident in how the Industrial Revolution from the 18th century enhanced societal development and growth in the gross domestic product (GDP) globally (Dogaru, 2020; International Energy Agency, 2022). In meeting electricity needs, fossil fuels have served as the primary means globally (Aboagye et al., 2021) with a proportion of approximately 80 % in the global energy mix (International Energy Agency, 2022). However, the adverse environmental consequences associated with its use, including the exhaustion of non-renewable energy resources, volatile pricing and climate change, compel the promotion of sustainable energy sources for global electricity production (Aboagye et al., 2021; IEA, 2015; Panwar et al., 2011).

Despite the challenges of identifying a scalable and reliable energy source with minimal environmental impact, the significance of hydropower (HP) remains undeniable. Compared to other energy sources, HP is recognised for its reliability and relatively lower ecological footprint (Kusre et al., 2010). With an estimated HP potential of nearly 500 GW (International Hydropower Association, 2020), Africa holds the greatest opportunity to harness HP for advancing the achievement of Sustainable Development Goal 7 (United Nations, 2015).

The lower carbon footprint of HP, in contrast with other energy sources, has significant contributions towards sustainable development and climate change mitigation (Allegretti et al., 2022; Lluís Gómez-Gener et al., 2023). Aside from the role of electrification, benefits such as controlling floods, serving as a sound water supply system and enhancing irrigation (Arthur et al., 2020), HP is also popular for its

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<https://doi.org/10.1016/j.esd.2025.101720>

Received 30 August 2024; Received in revised form 17 March 2025; Accepted 24 March 2025

Available online 1 April 2025

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widespread availability and long lifespan (Guamel & Lee, 2020). This underscores the significant potential of HP to drive socio-economic advancement. Although large HP dams offer these important benefits, their construction is associated with substantial environmental and social consequences, including biodiversity loss, greenhouse gas emissions (CO₂ and CH₄), population displacement, and disruptions to livelihoods (Arantes et al., 2023; Kuriakose et al., 2022; Paish, 2002).

The bid to maximise the use of HP on a large scale as the most preferred low-carbon option is also hampered by cost (Arthur et al., 2020), specifically for developing nations. SHP systems are recognised for their cost-effectiveness due to several key factors. While the initial investment in SHP can be substantial, the operational expenses are significantly lower than those of thermal and nuclear alternatives (Kalitsi, 2003). This is primarily because SHP plants do not require fuel, reducing ongoing costs associated with energy production (Duque et al., 2016; Jackson III & Cárdenas-Lailhacar, 2010). Additionally, SHP facilities typically have longer lifespans and require less complex maintenance compared to other renewable energy sources like wind power, which can suffer from mechanical degradation over time (Jackson III & Cárdenas-Lailhacar, 2010; Ugwu et al., 2022). Furthermore, SHP projects often benefit from faster payback periods due to their ability to generate a consistent and reliable power supply, which is particularly valuable in remote or off-grid locations (Sedona Investments, 2023). The cost-effectiveness of SHP is also enhanced by its potential for autonomy, allowing communities and businesses to reduce their reliance on external energy sources and mitigate the financial risks associated with fluctuating fossil fuel prices (Sedona Investments, 2023). Compared to large hydropower projects, SHP systems often have lower initial costs per unit of installed capacity, making them more accessible for smaller-scale investments (Guamel & Lee, 2020). This accessibility, combined with the environmental benefits and reliability of SHP, underscores its importance as a cost-effective and sustainable energy solution for developing countries (Sedona Investments, 2023). Due to its cost-effectiveness, SHP continues to serve as the fundamental pillar of the energy economy in several nations like China, Brazil and India (Yankey et al., 2023). Hence, SHP systems are crucial for sustainable development.

Ghana possesses significant water resources that could be effectively utilised sustainably to fulfil the increasing demand of its population. Ghana experiences an annual water runoff of approximately 56.4 billion cubic metres (Yeleliere et al., 2018). Therefore, implementing measures to harness and utilise this abundant water resource would represent a significant advancement. Before the end of 1998, most of Ghana's electrical power originated from a pair of hydroelectric dams located in Kpong and Akosombo with a total generation capacity of 1072 MW (Kalitsi, 2003). More recently, the 400 MW Bui and 45 kW Tsatsadu plants have also contributed to the country's energy supply (Bui Power Authority, 2019).

However, over the years, thermal power has become the more dominant source of electricity in Ghana, with an installed capacity of over 3700 MW (Ghana Investment Promotion Centre, 2022). As stated by Essel (2015), Ghana has approximately "17 medium and 22 mini/small HP sites with exploitable capacities ranging from 15 kW to 100 MW". These sites collectively have an estimated exploitable capacity of 800 MW (Aboagye et al., 2021). Considering all these findings, exploiting these potential HP sites is imperative to augment Ghana's energy mix. Although Ghana is recognised as a leader in electricity access in Sub-Saharan Africa, boasting an impressive rate of approximately 88.54 % (Ministry of Energy, 2022), it is hinged on significant energy sector debt (Atuahene & Sheng, 2023). This situation is mainly due to the prevalence of power generation from thermal sources. The dominance of thermal power generation not only exposes the country to financial vulnerability due to exorbitant fuel and maintenance costs (Energy Commission, 2019), but also to the emission of greenhouse gases (GHGs) at unusual levels. Cost is also a significant barrier to grid extension in rural areas where the industrialisation agenda can be

pushed to enhance socio-economic development.

Despite its abundant water resources, Ghana has yet to harness its potential for SHP development fully (Kabo-Bah et al., 2018). The country currently operates only one micro hydroelectric facility, specifically the 45 kW Tsatsadu micro power plant, developed by the Bui Power Authority after more than a decade of planning and implementation (Bui Power Authority, 2019). The limited expansion of SHP in Ghana is attributed to several challenges, including data paucity, limited technical expertise, inadequate infrastructure, and environmental concerns (Dernedde & Ofosu-Ahenkorah, 2002).

Under the seventh goal of the SDGs, which strives to guarantee accessible and sustainable energy for all (United Nations, 2015), it is imperative to harness various energy sources to fulfil the rising demand. This strategy involves exploring RE options, mainly HP. In tandem with the SDGs, Ghana's Renewable Energy Master Plan (REMP) aims to achieve >1000 MW in its RE portfolio by 2030 (Ministry of Energy, 2019). However, only approximately 132 MW of RE sources have been integrated, signifying a 2.40 % progress including embedded generation capacity (Energy Commission, 2024). Meeting rural communities and regions remotely displaced from the grid as a rural development initiative, particularly with SHP systems, is crucial since Ghana lags in its RE targets (Aboagye et al., 2021). Small-scale hydro generation in areas with HP potential can boost socio-economic activity (Arnaiz et al., 2018) and drive Ghana closer to meeting its RE targets. Additionally, SHP can significantly contribute to the net zero carbon initiative and mitigate the deterioration of the environment. Integrating these projects aligns with global climate commitments, such as the Paris Agreement, and supports the advancement of Ghana's Nationally Determined Contributions (NDCs) (Nyasapoh et al., 2022).

Hydrological modelling is crucial in assessing water resources and potential HP sites. For instance, Fasipe et al. (2021) used the Natural Resources Conservation Service – Curve Number (NRCS-CN) method to simulate hydrological characteristics in the Osse Sub-Basin, highlighting its simplicity and dependability for ungauged basins. Similarly, Areri and Bibi (2023) employed the Hydrologic Engineering Centre – Hydrologic Modelling System (HEC-HMS) to analyse rainfall-runoff relationships in the Genale Dawa basin, demonstrating the importance of hydrological modelling in understanding basin dynamics.

The development of SHP is vital for enhancing RE contributions in West Africa, particularly within the Economic Community of West African States (ECOWAS), which aims to increase the share of renewable energy to 19 % by 2030 (ECREEE, 2015). In Ghana, SHP has been identified as a promising source of clean energy. Studies such as those by Arthur et al. (2020) and Yankey et al. (2023) have utilised the SWAT model to evaluate dam siting and propose cascade systems for selected hydropower locations. Similarly, Agyemang-Boakye et al. (2024) conducted economic analyses of proposed dam and run-of-river (RoR) schemes using RETScreen.

Beyond Ghana, other regions have also seen significant efforts in SHP assessment. Kouadio et al. (2022) integrated SWAT with the Quantum Geographic Information System (QGIS) to assess hydropower potential in the White Bandama Watershed (Côte d'Ivoire). They used geospatial data to characterise the watershed hydrologically and hydro-geomorphologically, validating model performance with metrics like the Nash–Sutcliffe efficiency coefficient and identifying 22 potential hydropower sites. Ayik et al. (2023) focused on RoR SHP potential in Central Equatoria State (South Sudan), using a digital elevation model and stream network to locate sites. They employed SWAT for discharge estimation and identified 82 potential sites with a total output power of 1651.69 MW.

Despite these focused efforts on theoretical assessments of SHP potential, there is a notable gap in empirical studies that comprehensively evaluate both economic viability and environmental impacts, particularly regarding GHG emissions. This gap limits our understanding of how SHP can effectively contribute to regional and national energy goals.

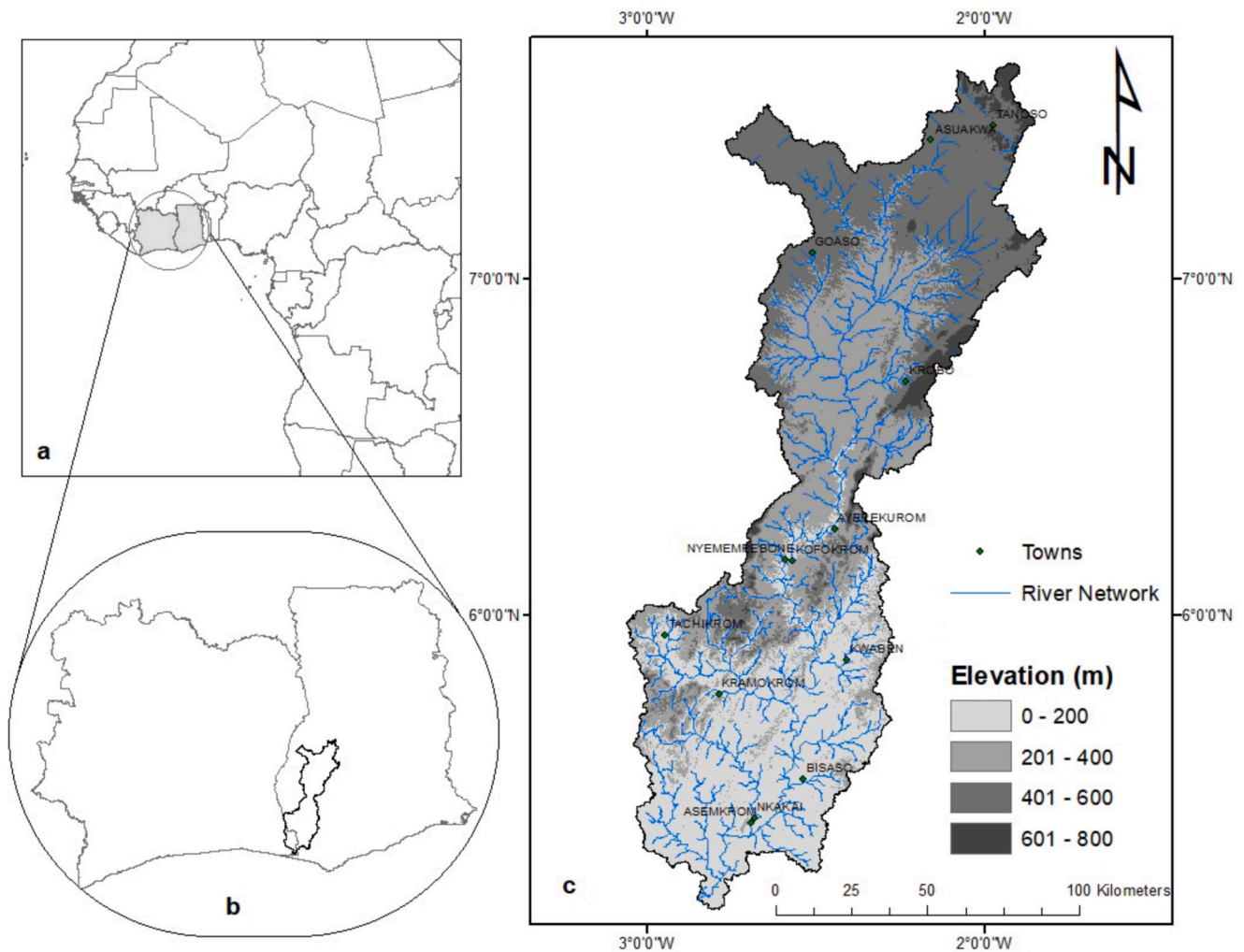


Fig. 1. Geographic position of the Tano Basin a) Africa b) Ghana and Ivory Coast c) Tano.

This study addresses this gap by focusing on the Tano Basin, employing the SWAT model for hydropower analysis and using RETScreen for techno-economic and emissions assessments. The specific contribution of this study is the integration of emissions analysis alongside economic evaluation, providing a holistic framework that aids policymakers and investors in making informed decisions about SHP development in the ECOWAS sub-region. By focusing on the Tano Basin, this research enhances local understanding and contributes to broader regional strategies for sustainable energy development.

The remainder of this paper is structured as follows: [Section 0](#) presents the materials and methods, including the study area, hydrological modelling using SWAT, and the techno-economic analysis performed with RETScreen. [Section 0](#) discusses the results, covering hydropower site selection, model performance evaluation, firm power estimation, and economic feasibility. Finally, [Section 0](#) provides conclusions and policy implications based on the findings and outlines recommendations for future research.

Materials and methods

Study area

The Tano Basin, as demonstrated in [Fig. 1](#), located in Ghana, is a significant river basin system in the southwestern area. It is positioned between latitudes 4°50'N and 7°50'N, as well as longitudes 1°50'W and

3°15'W. With a catchment area of approximately 15,000 km², it covers around 35 % of Bono, Bono East and Ahafo, 15 % of Ashanti, and 50 % of Western Regions. The primary Tano River originates from the highlands near Tuobodom, close to Techiman in the Bono East Region, at an elevation of 518 m above sea level ([Water Resources Commission, 2012](#)). It flows for 400 km before merging with the Aby Lagoon in Côte d'Ivoire. The TRB experiences a sub-equatorial climate characterised by two primary rainy periods, one from May to July and the other from September to November. The spatial distribution of annual rainfall accumulation from 1985 to 2016 varies from 1100 mm in the north to 1800 mm in the south, with an average value of approximately 1390 mm ([Larbi et al., 2022](#)). Rainfall distribution intensifies as one moves southward, and the annual relative humidity ranges between 75 % and 85 %. On average, there are between 90 and 140 days of rainfall throughout the year ([Nasirudeen et al., 2021](#); [Water Resources Commission, 2012](#)).

The Soil and Water Assessment Tool (SWAT) model and flow estimation

SWAT is a hydrological model that has been used around the globe ([Bahati et al., 2021](#); [Dallison et al., 2021](#); [Guo et al., 2021](#); [Larbi et al., 2022](#)). It is a watershed scale model that functions continuously on a daily time interval ([Yankey et al., 2023](#)). It has been extensively applied and can simulate long-term periods ([Arnold et al., 2012](#); [Solomon et al., 2024](#)). The SWAT model incorporates key components such as

Table 1
Dataset for SWAT modelling.

Data set	Source
Land Cover Map	Water Productivity Open-access portal https://data.apps.fao.org/wapor/?lang=en Date Accessed: 15th August 2023
Digital Elevation Model	United States Geological Survey Website https://earthexplorer.usgs.gov/ Date Accessed: 10th August 2023
Stream flow Data	Hydrological Services Department https://hydro.gov.gh/ Date Accessed: 1st August 2023
Climate (Temperature, Precipitation, Humidity, Solar Radiation and Wind Speed)	National Aeronautics and Space Administration (NASA) Power https://power.larc.nasa.gov/ Date Accessed: 31st August 2023
Soil Map	Food and Agriculture Organization https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/faunesco-soil-map-of-the-world/en/ Date Accessed: 25th August 2023

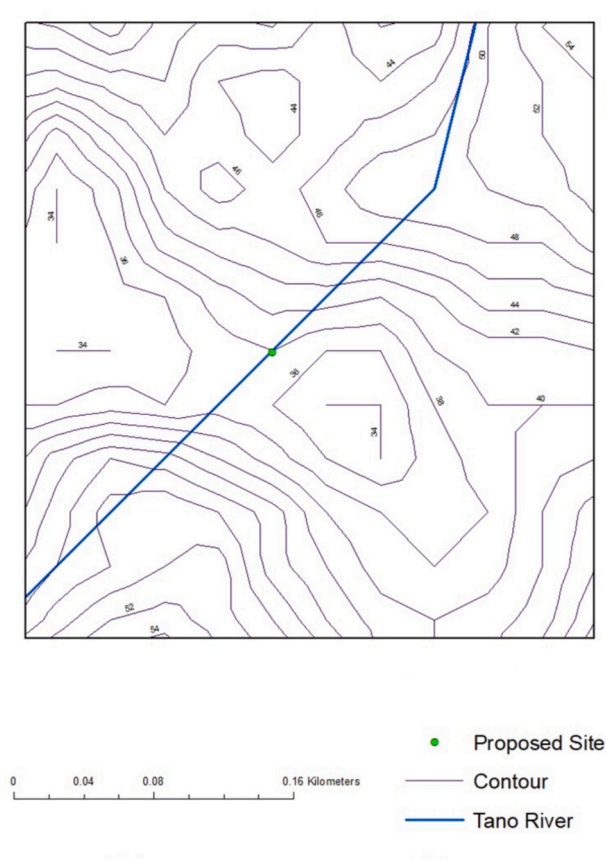


Fig. 2. Site selection

hydrology, soil properties, temperature, slope, weather and land use.

QSWAT, the SWAT extension in the QGIS interface, was used to simulate the streamflow at the proposed HP sites. QSWAT simulated flows using the spatial datasets outlined in Table 1. QSWAT divides basins into sub-basins, which are further divided into Hydrologic Response Units (HRUs). HRUs represent units with homogeneous soil characteristics, land use, and topology. In the SWAT simulation, HRUs may not be spatially contiguous but are expressed as a percentage of the sub-basin area (Arnold et al., 2012).

Water balance is the fundamental driving force behind all the

processes within SWAT. The simulation of watershed hydrology in SWAT consists of several phases, namely the land phase, the sediment and nutrient transport phase, and the water or routing phase of the hydrologic cycle (Arnold et al., 2012; Neitsch et al., 2002). The hydrologic processes simulated by SWAT encompass surface runoff, evapotranspiration, lateral flow, infiltration, canopy storage, and other relevant factors. Table 1 details the required SWAT input data sourced from diverse origins.

In SWAT, the hydrologic cycle is simulated using the water balance equation, which can be represented as follows:

$$S_{wt} = SW_o + \sum_{i=1}^t (R_{day-i} - Q_{surf} - E_a - W_{seep} - Q_{gw}) \quad (1)$$

where S_{wt} is the final soil water content (mmH_2O), SW_o is the initial soil water content on day i , t is the time in days, R_{day-i} is the amount of precipitation on day i (mmH_2O), Q_{surf} is the amount of surface runoff on day i (mmH_2O), E_a is the amount of evapotranspiration on day i (mmH_2O), W_{seep} is the amount of water entering the vadose zone from the soil profile on day i (mmH_2O) and Q_{gw} is the amount of return flow on day i (mmH_2O).

Digital Elevation Model (DEM) pre-processing

A DEM is crucial for accurately modelling the topography of the Tano River Basin, which is essential for simulating water flow and potential energy production. A 30-m Shuttle Radar Topography Mission Digital Elevation Model (SRTM-DEM) was acquired from the United States Geological Survey (USGS) due to its high resolution and global coverage, allowing for precise calculations of basin parameters. To ensure data accuracy, the pre-processing phase of the DEM involved sink filling from the Fill tool in QGIS to mitigate resolution inaccuracies and rounding errors. The flow direction and flow accumulation tools in the processing toolbox of QGIS were then used to determine the flow direction by estimating the flow from each raster cell to downstream cells, subsequently accumulating the flow. The Strahler function in the Processing Toolbox of QGIS was employed to generate the stream network via the resulting flow accumulation map.

The flow volume dictates the positioning of feasible locations based on an appropriate elevation difference. According to the Strahler stream order, higher-order streams were preferred as they suggested a greater water flow. The Strahler method, chosen for its simplicity and widespread use (Arthur et al., 2020; Kusre et al., 2010; Yankey et al., 2023), organised the stream network without requiring the mainstream traces back to its origins. This ensured that order four and above streams were selected to guarantee sufficient flow. The combination of flow and head influenced the hydropower generated, chosen with references to the works of Agyemang-Boakye et al. (2024), Arthur et al. (2020) and Yankey et al. (2023).

The Strahler approach categorises stream origin as first-order, with the convergence of two first-order streams forming a second-order stream, and this pattern continues progressively. A stream of higher-order is formed when it crosses a lower-order stream.

Availability of adequate head and flow for candidate RoR sites

The final site choice focused on 5th and 4th order streams to ensure sufficient flow. Intake points were strategically placed on stream orders 5 and 4, selected from below the watershed. A map of contours with intervals of 2 m was created using the DEM within the QGIS platform. The number of contour lines intersecting a segment of the stream and the variation in elevation (head) was used to ascertain the gross head measurement. To ensure compliance with the powerhouse location criteria, candidate sites needed a minimum gross head of 10 m to the Surge Tank. This criterion is aimed at positioning the Surge Tank on land, distanced from the river, and close to the anticipated power

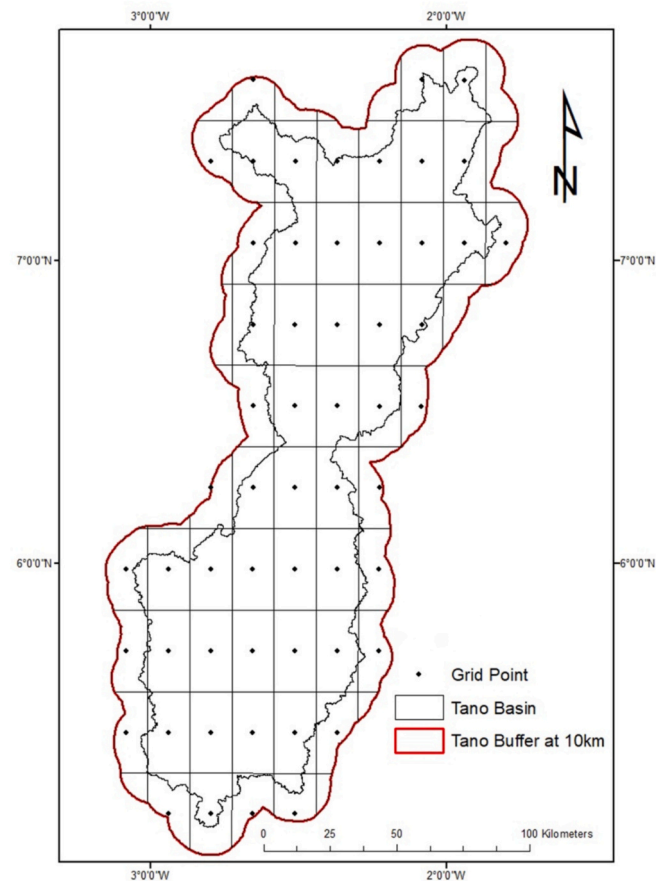


Fig. 3. Figure showing grid points on the Tano basin with a buffer of 10 km.

Table 2
Model parameters and corresponding spatial coverage.

Model parameter	No. of grid points/stations	Year
Temperature	39	1992–2022
Rainfall	39	1992–2022
Stream flow	1	2003–2018

generation site (Alcalá et al., 2021). Fig. 2 shows the site selection process in QGIS.

Spacing and siting of hydropower

Sites for HP potential on the Tano River were chosen according to the criteria adopted by Agyemang-Boakye et al. (2024). The relatively level terrain (mainly the southern part) of the Tano Basin led to the adoption of a distance of at least 1 km to ensure ample space between two subsequent candidate sites. This distance is not only to foster ecosystem regeneration but also to accommodate various HP components, such as the penstock. To identify appropriate locations for extracting climate data from the National Aeronautics and Space Administration (NASA) Langley Research Center (LaRC) POWER database, the fishnet tool within the Data Management toolbox in QGIS was employed, as illustrated in Fig. 3. This step was crucial for establishing a grid of evenly spaced points across the basin, ensuring precise data extraction. Table 2 provides the number of stations, or grid points utilised in the modelling for each parameter. Following the approach of Ayik et al. (2023), a 10-kilometre buffer was applied around the Tano Basin to ensure the inclusion of rivers intersecting its boundaries.

SWAT project setup and data preparation

The SWAT model's setup followed a structured six-stage process, beginning with project naming and configuration. Each stage was sequentially dependent, requiring the successful completion of the preceding step before advancing. Essential climate data, including daily minimum and maximum temperatures and precipitation, were collected and formatted to ensure compatibility with the model. Streamflow data, recorded daily, was also incorporated for calibration and validation purposes.

To ensure spatial consistency, all geospatial datasets, including soil, land use/land cover (LULC), and DEM maps, were reprojected from the geographic coordinate system (GCS) to the Universal Transverse Mercator (UTM) coordinate system (WGS 1984, UTM Zone 30). This reprojection was necessary to align spatial data for accurate hydrological simulations within SWAT (Winchell et al., 2010).

The delineation of the watershed

Following the project setup, the watershed delineation process was conducted, establishing the stream network using the DEM. QSWAT was employed to compute flow direction, flow accumulation, and stream networks, with an automatic sink-filling procedure applied to correct depressions and ensure hydrologic continuity. Flow accumulation was determined using the raster calculator, considering only cells with an accumulated flow of at least 10,000 (Arthur et al., 2020). The delineation process incorporated key outlets, including identified locations, stream gauging stations, and the proposed SHP sites. Based on these outlets, the watershed was subdivided into 31 sub-basins, forming the basis for subsequent hydrological modelling.

Analysis of HRUs and SWAT simulation

In the SWAT model, sub-basins are further divided into HRUs, which represent unique combinations of land use, slope, and soil characteristics. While HRUs are not directly observable, they serve as computational entities within SWAT simulations, allowing for a more detailed representation of watershed processes. A hierarchical classification system was used to define HRUs, categorising land surface slope into three classes: 0–10 %, 10–20 %, and 20–99.99 %. The classification thresholds were set at 5 % for soil type and 10 % for land use and slope, ensuring the inclusion of dominant classes in HRU delineation (Her et al., 2015). A total of 244 HRUs were generated and distributed across the predefined slope categories.

The land cover map and its classification reference table were imported into QSWAT, after which LULC data were reclassified to match the specified categories. Similarly, soil classification was adjusted based on user-defined criteria, and the slope was reclassified accordingly. These layers were then overlaid to generate the HRU feature class for the watershed. The slope data was crucial in determining dominant LULC and soil types.

Climate data, including daily minimum and maximum temperature, precipitation, solar radiation, wind speed, and relative humidity, were integrated into the model. After incorporating essential inputs such as elevation, station locations, and station names, the SWAT model was initiated on January 1, 1992, with a three-year warm-up period and a simulation end date of December 31, 2022. Following the initial run, model calibration was performed using manual and automatic approaches. Manual calibration within the QSWAT interface was employed to identify the most sensitive parameters for the basin before further refinement.

Model calibration and validation

While the data inputs were initially recorded on a daily time scale, the calibration and validation of the SWAT model were performed on a

Table 3
Parameter values for SWAT model simulation.

Parameters	Description	Initial range	Fitted value
r_CN2.mgt	Curve number	−0.5 to 0.5	0.19
v_ALPHA_BF. gw	Baseflow alpha factor	0 to 1	0.74
v_GW_DELAY. gw	Groundwater delay-time	0 to 500	52.05
v_GWQMN.gw	Groundwater minimum threshold	0 to 5000	0.39
r_SOL_AWC(). sol	Saturated hydraulic conductivity	−0.5 to 0.5	0.12
r_ESCO.hru	Soil evaporation compensation factor	−0.1 to 0.1	0.32
r_GW_REVAP. gw	Groundwater revap coefficient	0.02 to 0.2	0.00
r_REVAPMN. gw	Threshold depth of water in the shallow aquifer for 'revap' to occur	0 to 500	0.08
r_CH_N2.rte	Mannings 'n' coefficient	−0.01 to 0.3	0.27
r_HRU_SLP.hru	Average slope steepness	0 to 1	0.02
r_OV_N.hru	Manning's "n" value for overland flow	0.01 to 1	−0.06
r_SLSUBBSN. hru	Average slope length	10 to 150	0.00

monthly time scale using streamflow data sourced from the Hydrological Services Department. Using the Sefwi Wiawso stream flow data, calibration and validation took place with calibration data spanning from January 1, 2003, to December 31, 2010. The model first underwent manual calibration. This was followed by automatic calibration employing the Sequential Uncertainty Fitting (SUFI-2) procedure in the SWAT Calibration and Uncertainty Program (SWAT-CUP). Following procedures proposed by [Neitsch et al. \(2005\)](#), manual calibration was employed using the QSWAT interface on a monthly time step. Model parameters were adjusted within their specified minimum and maximum limits, with each parameter having a threshold guiding the manual calibration process. The model parameters selected for the automatic calibration were CN2, GWQMN, SOL_AWC, ESCO, GW_REVAP, OV_N, REVAPMN, GW_DELAY, HRU_SLP_ALPHA_BF, SLSUBBSN and CH_N2. Adjustments were made by either multiplying by a factor (r), adding an increment (a) or replacing (v). These adjustments were made first to variables influencing surface runoff, followed by those defining lateral and groundwater flow ([Agyemang-Boakye et al., 2024](#)). Model output efficacy was evaluated using NSE, R^2 , RSR, R-factor, P-factor, and PBIAS criteria. Desirable efficiencies of various model parameters that were not initially achieved were further adjusted until satisfactory results were obtained. Adjustments ceased when the model

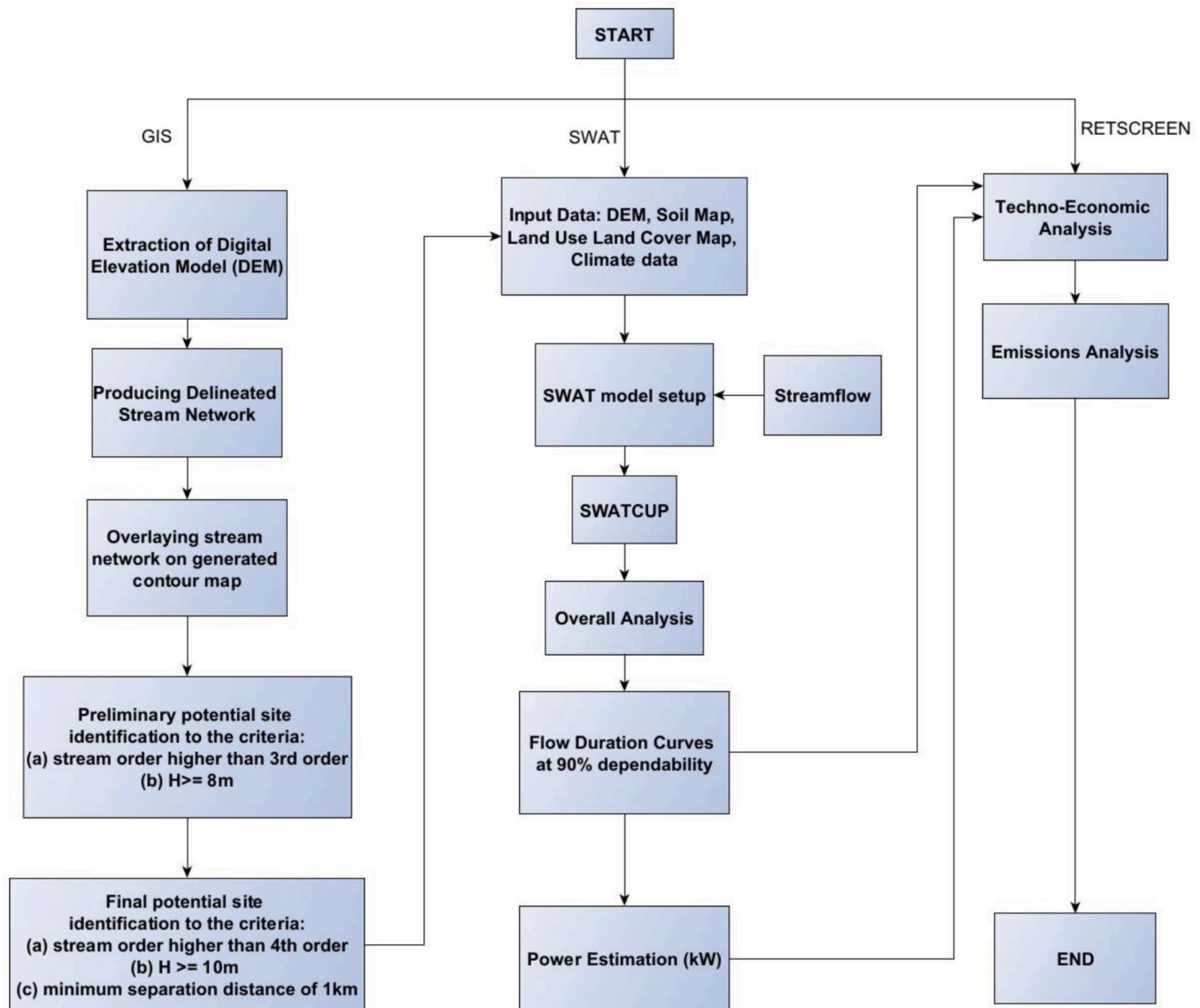


Fig. 4. Methodological framework for the study.

Table 4
Hydro formula costing method.

Parameter	Unit	Value
Local vs. Canadian equipment cost ratio		1.00
Local vs. Canadian fuel cost ratio		0.93
Local vs. Canadian labour cost ratio		0.133
Equipment manufacture cost coefficient		1.00
Exchange rate	\$/CAD	1.3661
Number of turbines	Turbine	2
Cold climate	Yes/No	No

output efficiency was deemed satisfactory without altering the previously adjusted parameters. The calibration and validation of the model involved identifying key parameters that significantly influenced the model's output, along with their respective descriptions, as detailed in Table 3. The decision to use data from 2003 to 2010 for calibration and from 2011 to 2018 for validation was based on standard practices in hydrological modelling, such as those used in SWAT model applications. Calibration typically involves adjusting model parameters to match observed data, while validation assesses the model's predictive capability using independent data (Gassman et al., 2007; Moriasi et al., 2007). The separation of data into these periods allowed for a robust evaluation of the model's performance under different conditions.

The authors chose these specific periods because they cover a range of hydrological conditions, including wet and dry years, which is crucial for ensuring that the model can accurately simulate various scenarios (Arnold et al., 2012). The calibration period (2003–2010) was selected to establish baseline model performance, while the validation period (2011–2018) was used to evaluate the model's ability to predict flows under different conditions without being influenced by the calibration process. Fig. 4 illustrates the methodological framework for this study.

Firm power and energy estimation

Following the calibration and validation of the model, monthly streamflow data were used to generate flow duration curves (FDCs) and power duration curves (PDCs). FDCs and PDCs are critical in hydropower development as they provide insights into the frequency and reliability of streamflow levels and their corresponding power output over a given period (Alakbar & Burgan, 2024; Vogel & Fennessey, 1994). Specifically, they illustrate the percentage of time that streamflow exceeds a specified discharge threshold, aiding in the assessment of water availability for power generation (Castellarin et al., 2013).

The model simulated sequential water discharge data from 2003 to 2018, aligning with the availability of climate data. The discharge data modelled by the SWAT model was used to construct an FDC for all candidate HP sites. Subsequently, the percentage dependability was calculated using the Eq. (2).

$$\%dependability = \frac{\text{Cumulative Number of Flows (N)}}{\text{Number of Data Set (n)}} * 100 \quad (2)$$

A 90 % dependability was employed in this study. A 90 % dependability is widely used in hydropower planning as it represents a conservative yet realistic estimate of water availability (Ridolfi et al., 2020). This level ensures that the designed power output can be reliably met or exceeded 90 % of the time. Moreover, a 90 % dependability allows for a more sustainable approach, ensuring sufficient water remains in the river for ecological needs even during drier periods. The corresponding flow for each site at this dependability level was identified. In the process of estimating power, a general efficiency level of 75 % was employed (Phitaksurachai et al., 2017). This efficiency rating of 75 % applied is influenced by factors, including hydraulic head losses, transformer efficiency, and turbine efficiency, which contribute to a reduction in the overall efficiency as established by Edeoja and Awuniji (2017). The potential electrical generation capacity of a hydro facility is expressed as:

$$P = \frac{\rho g Q_{90} H \eta}{1000} \quad (3)$$

where, P is the output of power (kW), ρ is the water density (1000 kg/m^3), g is the acceleration due to gravity, Q_{90} is the discharge at the proposed location corresponding to 90 % dependability ($\frac{\text{m}^3}{\text{s}}$), H is the head (m) at the proposed location and η is the overall efficiency of the project.

For this study, with an overall efficiency of 75 % (Agyemang-Boakye et al., 2024) and a focus on 90 % flow dependability, the power output from a SHP was estimated as:

$$P = 7.35 Q_{90} H \quad (4)$$

The estimation of firm energy involved a plant capacity factor C_f of 60 %, exemplified by the research conducted by Kusre et al. (2010) and Reichl and Hack (2017). To ensure a continuous power supply during periods of available flow, the proposed locations were expected to operate for 328 days out of 365 days annually.

$$\text{Energy (kWhr)} = P T_o C_f \quad (5)$$

where P is the power (kW), T_o is the time in hours and C_f is the plant factor.

Techno-economic analysis with RETScreen

A critical part of the assessment of SHP projects is their feasibility and sustainability in techno-economic analysis. This stage studies the intricate relationship between technology, engineering, and financial viability as a requirement to judge whether a project is to be cleared for SHP development (Singal et al., 2010). Potential investors need a comprehensive understanding of key cost factors to make well-informed investment decisions in small-scale HP projects. These factors include initial capital investments, ongoing operational expenses, expected lifespan, and utilisation rate (IRENA, 2012). Consideration was given to external factors, including the discount rate and the overall cost of capital. Several financial performance metrics were analysed to assess projects from an economic perspective, including internal rate of return (IRR), payback period and net present value (NPV).

There are two approaches to project costing in the Small Hydro Project model of RETScreen: the hydro formula and the detailed costing methods. The detailed costing method allows users to make cost estimates based on estimated quantities and unit costs. This method requires that users provide detailed and site-specific information, including the size and layout of specific structures. The hydro formula costing method, on the other hand, relies on empirical cost equations derived from over 20 years of data collected from both small and large hydro projects (RetScreen Expert, 2020). The hydro formula costing method was chosen for its simplicity, empirical basis, and general applicability, making it ideal for initial assessments and comparisons across different sites. The hydro formula costing formula can be accessed under the Cost tab in the Hydropower Project Model. By default, the formula costing method uses Canadian projects as a benchmark, allowing users to switch the results to their respective locality (Ghana was used in this scenario). The deficiency of a hydro manufacturing industry in Ghana necessitated that the relative cost of equipment and equipment manufacturing be based on the manufacturing industry in Canada. At the same time, the price of fuel and labour was centred on Ghana and Canada, factoring in their respective ratios. The ratio of fuel and labour costs between Ghana and Canada was assessed for the 2023/2024 year, leading to the following findings:

Ghana's fuel (diesel) price was approximately 1.43 Canadian dollars (CAD)/Litre, and the Canadian fuel price was approximately 1.53 CAD/Litre (The values were used under an approximate exchange rate of 1 (US dollar) USD = 1.3361 CAD at the time of the assessment) (National

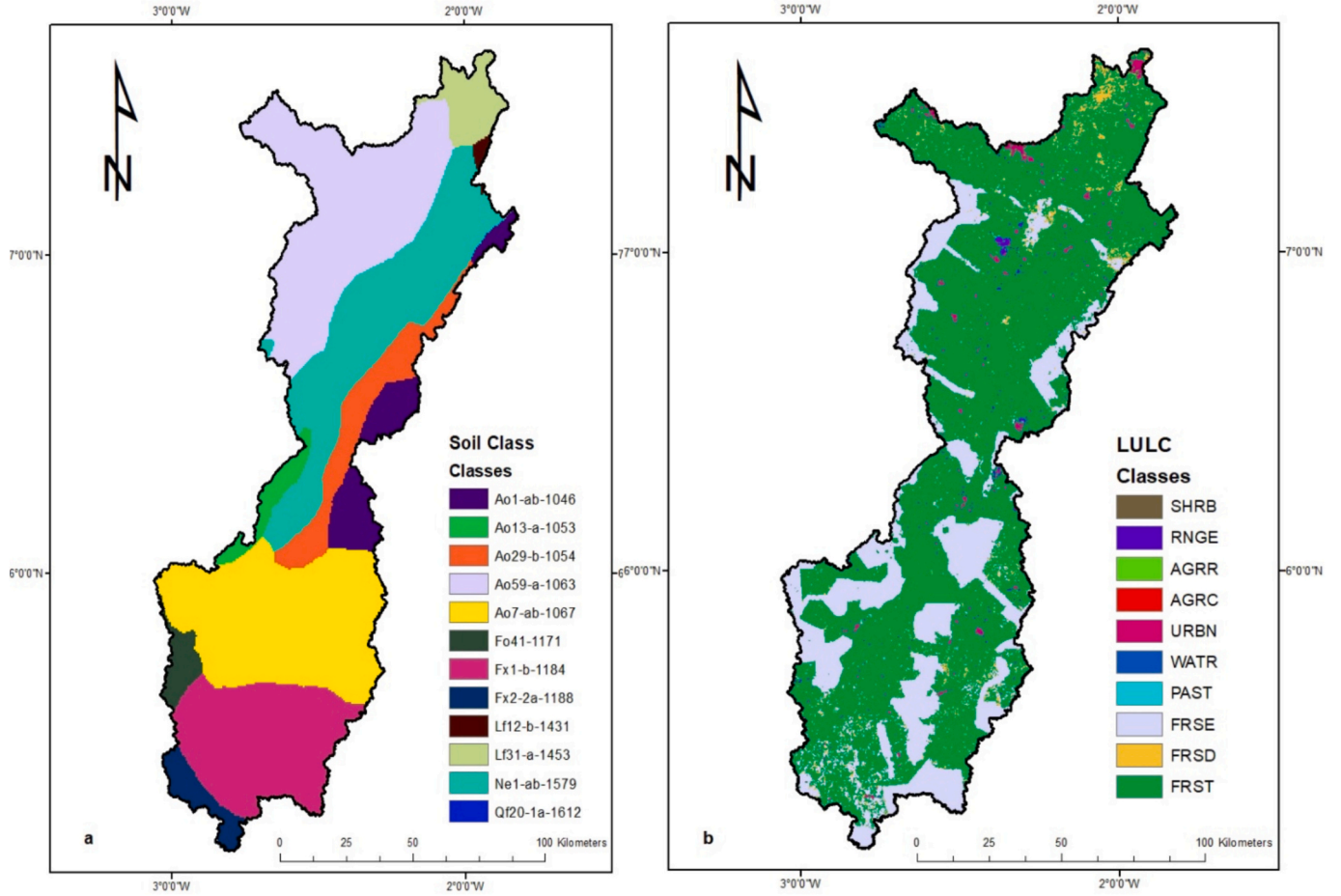


Fig. 5. Tano Basin a) soil b) LULC.

Petroleum Authority, 2024; Sasu, 2024). Therefore, Ghana versus Canada's fuel cost ratio was 0.93. Utilising the minimum wages in both countries, Canada's labour cost was found to be CAD 2973.76/month while Ghana's labour cost was GHC18.15 per day (CAD 397.57/month) (Ankrah, 2024; Retail Council of Canada, 2024). Ghana's labour cost ratio to Canada's was 0.133. The Ghanaian versus Canadian equipment cost ratio and the equipment manufacture cost coefficient were maintained at 1.00. Table 4 outlines the Hydro Formula Costing parameters for all the potential hydropower sites.

Net present value (NPV)

The NPV evaluates the present discounting of all future cash flows. This is given by:

$$NPV = \sum_{n=0}^N \frac{C_n}{(1+r)^n} \quad (6)$$

where C_n is cash flow for the year n , N is the project life in years, and r is the discount rate.

Internal rate of return (IRR)

The IRR is the discount rate at which the NPV is zero. It is given by the formula:

$$0 = \sum_{n=0}^N \frac{C_n}{(1+IRR)^n} \quad (7)$$

where N is the project life in years, C_n is cash flow for the year n (note that C_0 is the project's equity without incentives and grants; this is the cash flow for year zero).

Benefit-Cost Ratio (BCR)

The Benefit-Cost (B-C) Ratio is the ratio of the total benefits (income and savings) to the total costs of a project and is given by:

$$\frac{NPV + (1-f)C}{(1-f)C} \quad (8)$$

where C is the total initial cost of the project, and f is the debt ratio.

Simple payback period

The simple payback period is the years taken for cash flow to equal the total investment (comprising the sum of debt and equity). Here, the cash flow does not consider debt payments. The simple payback period formula is given by:

$$SP = \frac{C - IG}{(C_{ener} + C_{capa} + C_{RE} + C_{GHG}) - (C_{O\&M} + C_{fuel})} \quad (9)$$

where C is the total initial cost of the project, IG is incentives and grants, C_{ener} is the annual energy savings or income, C_{capa} is the annual capacity savings or income, C_{RE} is annual renewable energy production, C_{GHG} is the GHG reduction income, $C_{O\&M}$ is the yearly operation and maintenance cost incurred by the project, and, C_{fuel} is the annual cost of fuel or electricity.

Built-in functions to determine financial metrics, emissions and technical feasibility made RETScreen the ideal fit for this study (Retscreen Expert, 2020). Asamoah et al. (2023) used the RETScreen Expert software to perform a techno-economic assessment of a proposed 50 MW utility-scale wind farm at four locations in Ghana. The evaluation aimed to achieve a roadmap for wind energy in Ghana. The study

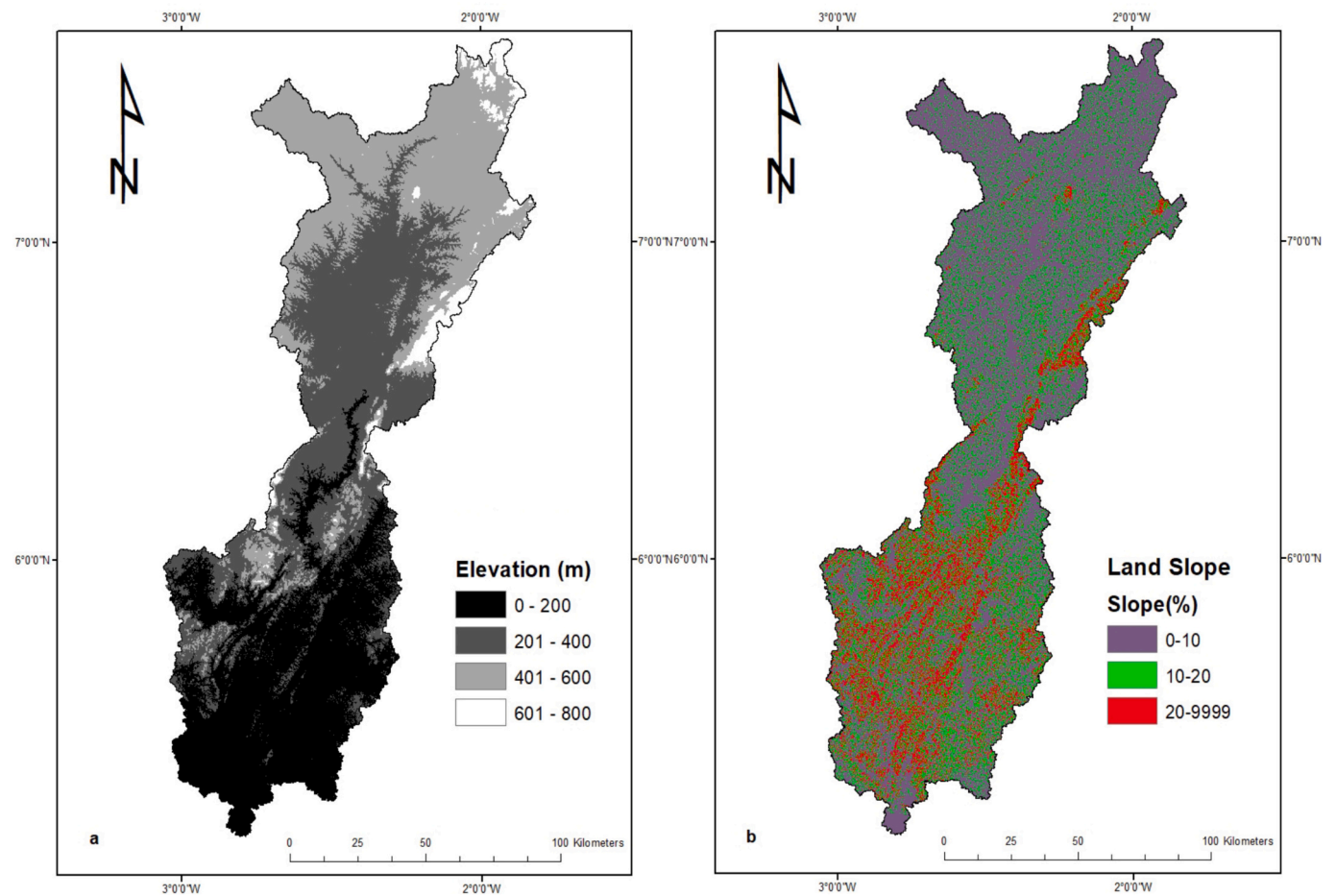


Fig. 6. Tano Basin a) elevation b) slope.

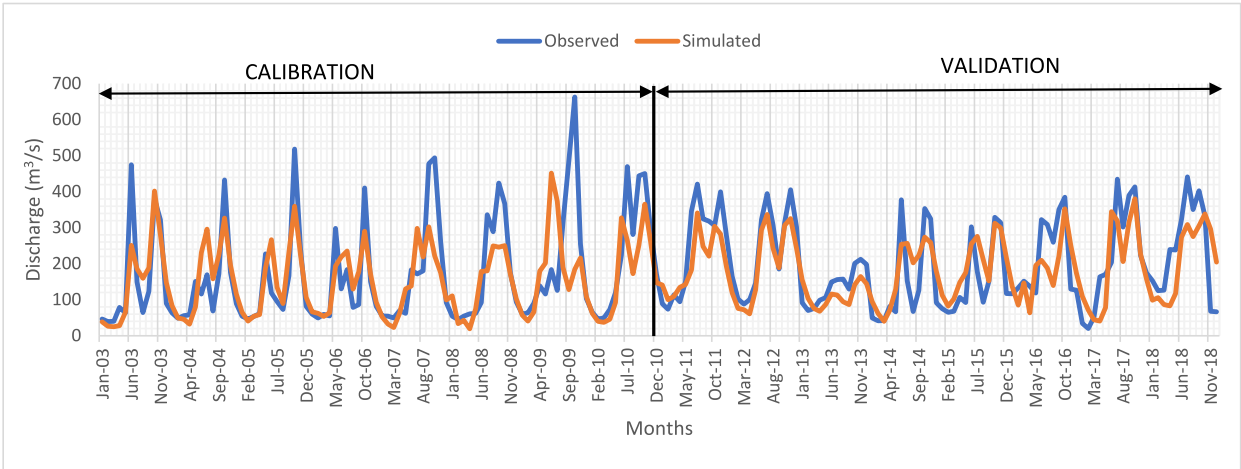


Fig. 7. Comparison between simulated and observed flows using SWAT.

revealed that all four sites produced favourable NPV results, indicating the financial feasibility of a potential wind farm project at those locations. In a similar vein, Akpahou et al. (2023) sought to assess the techno-economic feasibility of a 10 MW utility-scale grid-tied solar photovoltaic system in seven cities in Benin using the RETScreen Expert software. The results of the study showed that providing incentives and subsidies could reduce the levelized cost of electricity (LCOE) and encourage more significant investment in solar photovoltaic (PV) systems in Benin. The project revealed a potential capacity factor of 15.1 %

and a performance ratio of 67.3 %.

Results and discussion

Hydropower siting

The analysis revealed that the upland areas of the basin exhibited elevations of up to 800 m, signifying a high altitude. However, areas leading up to the bottomlands of the basin are characterised by steep

Table 5
Criteria for model performance.

Criteria	Calibration	Validation	Performance range	Performance ranking	Reference
NSE	0.45	0.58	$0 < \text{NSE} < 0.5$	Acceptable	(Moriassi et al., 2007; Yilmaz & Onoz, 2020)
R^2	0.47	0.60	$0.5 < \text{NSE} < 0.65$	Satisfactory	(Moriassi et al., 2007)
RSR	0.74	0.65	$R^2 > 0.5$	Acceptable	(Moriassi et al., 2007)
P-factor	0.45	0.48	$\text{RSR} < 0.7$	Satisfactory	(Abbaspour, 2015; Abbaspour et al., 2004)
R-factor	0.70	0.89	$0.4 < \text{p-factor} < 0.5$	Acceptable	(Abbaspour, 2015; Abbaspour et al., 2004)
PBIAS	8.9	8.2	$\text{r-factor} < 1$	Satisfactory	(Moriassi et al., 2007)
			$\text{PBAIS} \leq 10$	Good	

Table 6
Comparison of model output with existing literature.

Basin parameter (mm)	SWAT model	Water Resources Commission
Rainfall	1428.3	1300–2000
Evapotranspiration (ET)	478.4	1120
Potential Evapotranspiration (PET)	946	1322

slopes, as illustrated in Fig. 5. Steep slopes are advantageous for HP generation because they enhance the potential energy and velocity of water flow. This improves the efficiency and cost of power production

through a shorter penstock length and reduced diverted reach of stream (Kumar & Katoch, 2015; Yüksel, 2007). Given that HP generation is contingent on flow and elevation differences, it is imperative to identify the accessible flow and head in the region to assess HP potential. Figs. 5 and 6 illustrate the Soil and LULC and the Elevation and Slope of the Tano basin, respectively, according to the SWAT model.

Fig. 5 categorises ten classifications of land use: shrubland (SHRB), range-grasses (RNGE), agricultural land-row crops (AGRR), agricultural land-close grown (AGRC), built-up areas (URBN), water bodies (WATR), pasture (PAST), forest-evergreen (FRSE), forest-deciduous (FRSD), and forest-mixed (FRST). Using the Strahler method, five stream orders were

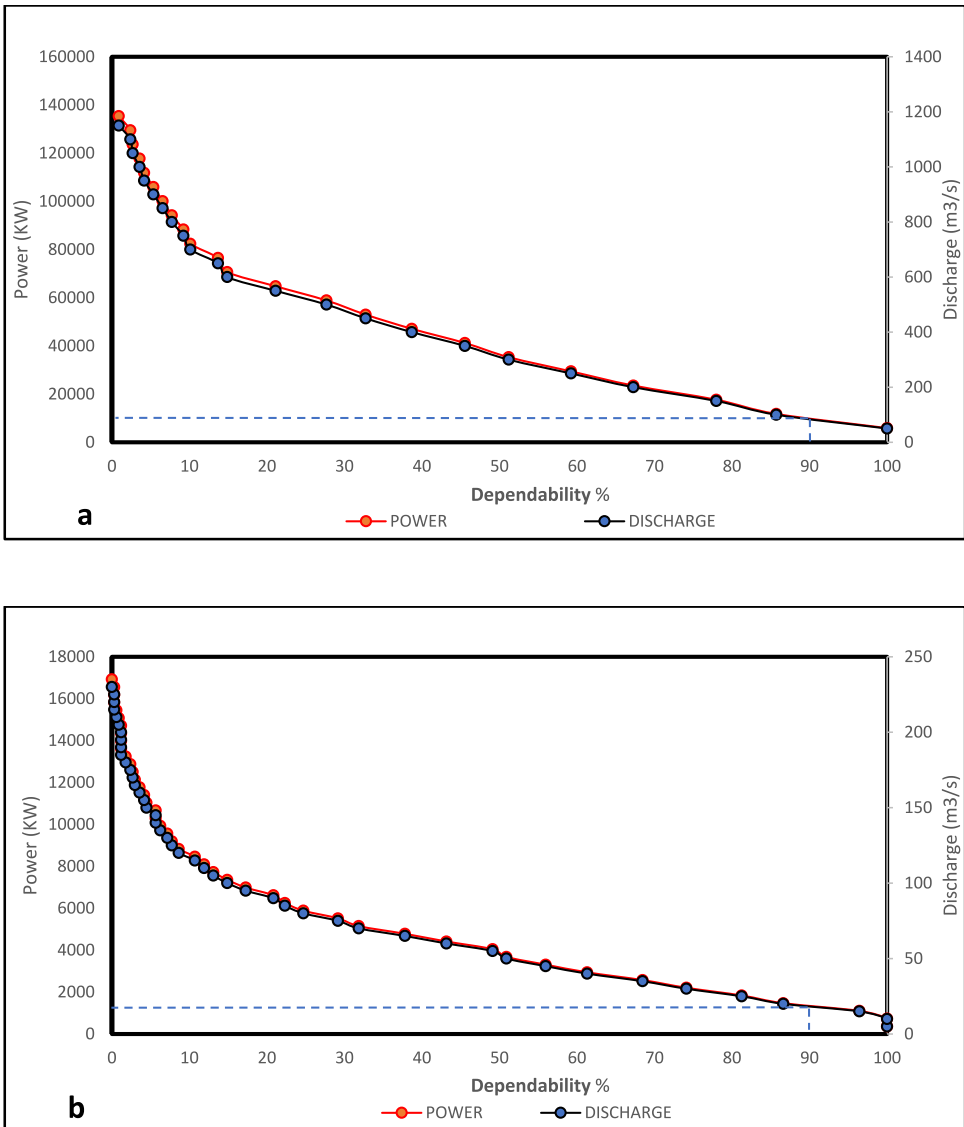


Fig. 8. FDC and PDC of hydropower sites a) HP 7 b) HP16.

Table 7

Firm power and energy at candidate sites

HP	Q ₉₀ (m ³ /s)	H (m)	η	Firm power (kW)	Firm energy (GWh)
1	96.94	10	0.75	7132.36	33.7
2	86.49	10	0.75	6363.50	30.1
3	86.28	14	0.75	8887.27	42
4	86.24	12	0.75	7614.13	36
5	86.13	10	0.75	6337.02	30
6	86.07	14	0.75	8865.64	41.9
7	83.45	16	0.75	9823.73	46.4
8	82.65	12	0.75	7287.17	34.5
9	77.29	16	0.75	9098.58	43
10	76.99	10	0.75	5664.54	26.8
11	76.84	16	0.75	9045.61	42.8
12	75.63	10	0.75	5564.48	26.3
13	75.39	14	0.75	7765.55	36.7
14	75.32	16	0.75	8866.67	41.9
15	74.89	10	0.75	5510.03	26
16	13.3	10	0.75	978.55	4.6
17	60.09	10	0.75	4421.12	20.9
18	59.57	14	0.75	6136.01	29
19	59.36	14	0.75	6114.38	28.9
20	59.36	10	0.75	4367.41	20.6
21	59.22	10	0.75	4357.11	20.6
22	59.2	10	0.75	4355.64	20.6
23	50.2	12	0.75	4416.27	20.9
24	48.61	10	0.75	3576.48	16.9
AVG	70.65	11.63	–	6582.92	28.8

Table 8

Details of the 24 proposed SHP sites.

Site	Head (m)	Latitude	Longitude	Stream reach (m)	Slope (%)	Stream order
1	10	5.4249	–2.61559	6724	8.9	5
2	10	5.43436	–2.60919	1378	10	5
3	14	5.45851	–2.60198	3382	10	5
4	12	5.46378	–2.59322	1769	39	5
5	10	5.47726	–2.60086	2199	10	5
6	14	5.54378	–2.5753	12,245	10	5
7	16	5.57107	–2.56616	3452	10	5
8	12	5.58224	–2.56503	1305	10	5
9	16	5.64461	–2.55448	3500	10	5
10	10	5.67669	–2.57281	5244	10	5
11	16	5.69557	–2.55503	5536	10	5
12	10	5.7046	–2.55752	5298	15	5
13	14	5.73576	–2.57675	1089	10	5
14	16	5.74971	–2.61786	4557	10	4
15	10	5.80794	–2.57001	7115	10	5
16	10	5.78655	–2.50559	24,178	7.4	5
17	10	5.79492	–2.50254	8668	13	5
18	14	5.8064	–2.50254	1079	10	5
19	14	5.83847	–2.51504	1260	10	5
20	10	5.8543	–2.51449	4186	10	5
21	10	5.92098	–2.5017	1820	10	5
22	10	6.05014	–2.58881	10,754	10	5
23	12	5.34742	–2.69364	24,776	15	5
24	10	5.61017	–2.57253	20,885	6.7	5

determined. The criteria of a stream order of 4 or greater and an appropriate head of not <8 m saw to the initial selection of 40 candidate sites.

The final selection of the RoR sites was guided by stricter criteria, including a minimum separation distance of 1 km between sites and intake locations, a minimum head of 10 m, and the condition that both sites and intake points be situated on stream orders 4 and 5. These stricter criteria were adopted to ensure adequate head and flow for sufficient power generation while eliminating suboptimal locations to prevent an excessive concentration of sites. This approach aligns with methodologies demonstrated by [Arthur et al. \(2020\)](#) and [Yankey et al. \(2023\)](#), which emphasise optimising hydropower site selection for technical and environmental feasibility.

SWAT model performance

[Fig. 7](#) depicts the analysed observed and simulated flows in the basin. The plot illustrates that the model predicts well, with little lag between the simulated and calibrated values between June 2009 and January 2010. The model significantly simulates the high peaks (representing extreme flow periods) and accurately simulates base flow periods. This signifies a model that performs satisfactorily and can be used for water resources management in the basin.

The primary criterion employed to assess the model's performance was the Nash-Sutcliffe Efficiency (NSE). Recording NSE values of 0.45 for calibration and 0.58 for validation, the model's performance was deemed satisfactory ([Moriassi et al., 2007](#); [Yilmaz & Onoz, 2020](#)).

Additional performance metrics, including R², RSR, and PBIAS ([Arnold et al., 2012](#); [Krause et al., 2005](#); [Moriassi et al., 2007](#); [Sittichok et al., 2018](#)) were also computed for calibration and validation periods, consistently supporting the model's overall adequacy. [Table 5](#) summarises these performance criteria.

Notably, the model revealed marginal underestimations of high flows, as indicated in [Fig. 7](#). Also, there were slight underestimations in low flows for certain periods relative to measured streamflow data. This may result from satellite precipitation data (instead of ground observations), which cannot accurately record extremely high or low rainfall. Satellite-derived reanalysis data (NASA POWER) were utilised to address gaps in the records from ground observations. While observed climate data, including precipitation, minimum temperature, and maximum temperature, were acquired from the Ghana Meteorological Agency (GMet), the data were inconsistent in terms of missing values in temporal coverage (1992–2022) and the required spatial coverage across the basin.

The results indicate that the NSE during the calibration period was lower than during the validation period. This discrepancy may be due to challenges in accurately simulating peak flows during calibration, possibly resulting from unaccounted-for processes within the subbasin. Additionally, the validation period appears to have experienced more consistent climatic conditions, which may have improved model performance by reducing variability in the input data. Due to the inherent prediction uncertainty in watershed modelling arising from its complexities, the R-factor and P-factor were considered to evaluate the model's competence in simulated discharge, reflecting recommended values for performance assessment ([Abbaspour, 2015](#)). After calibration and validation, the modelled flows were exported to develop FDCs and estimate firm power at the respective candidate sites.

The model's estimation indicated an annual rainfall of 1428.3 mm, which aligns with the 1300 to 2000 mm range reported in the Tano Basin's Integrated Water Resources Management (IWRM) Plan by the [Water Resources Commission \(WRC\) \(2012\)](#). [Table 6](#) compares the SWAT model's output parameters with the WRC ones.

The report from the WRC in 2012 represents an average of hydrological conditions annually. The SWAT model provides a reliable estimate of annual rainfall for the basin and aligns with findings from the study done by [Larbi et al. \(2022\)](#). Also, the model's ET estimate (478.4 mm) is significantly lower than the WRC's value (1120 mm), suggesting that the model either underestimates actual ET or the WRC's value might include additional factors not captured by the model. The PET estimated by the SWAT model (946 mm) is also lower than the WRC's reported value (1322 mm). This discrepancy could be due to differences in the methodologies or assumptions used in calculating PET.

Energy assessment

The FDC and PDC, illustrating flow conditions and power output at a 90 % dependability level, are presented in [Fig. 8](#) for two proposed locations (HP7 and HP16). These curves confirm the perennial nature of the Tano River, indicating consistent year-round flow. Constructing FDCs for each proposed site was crucial for accurately assessing HP

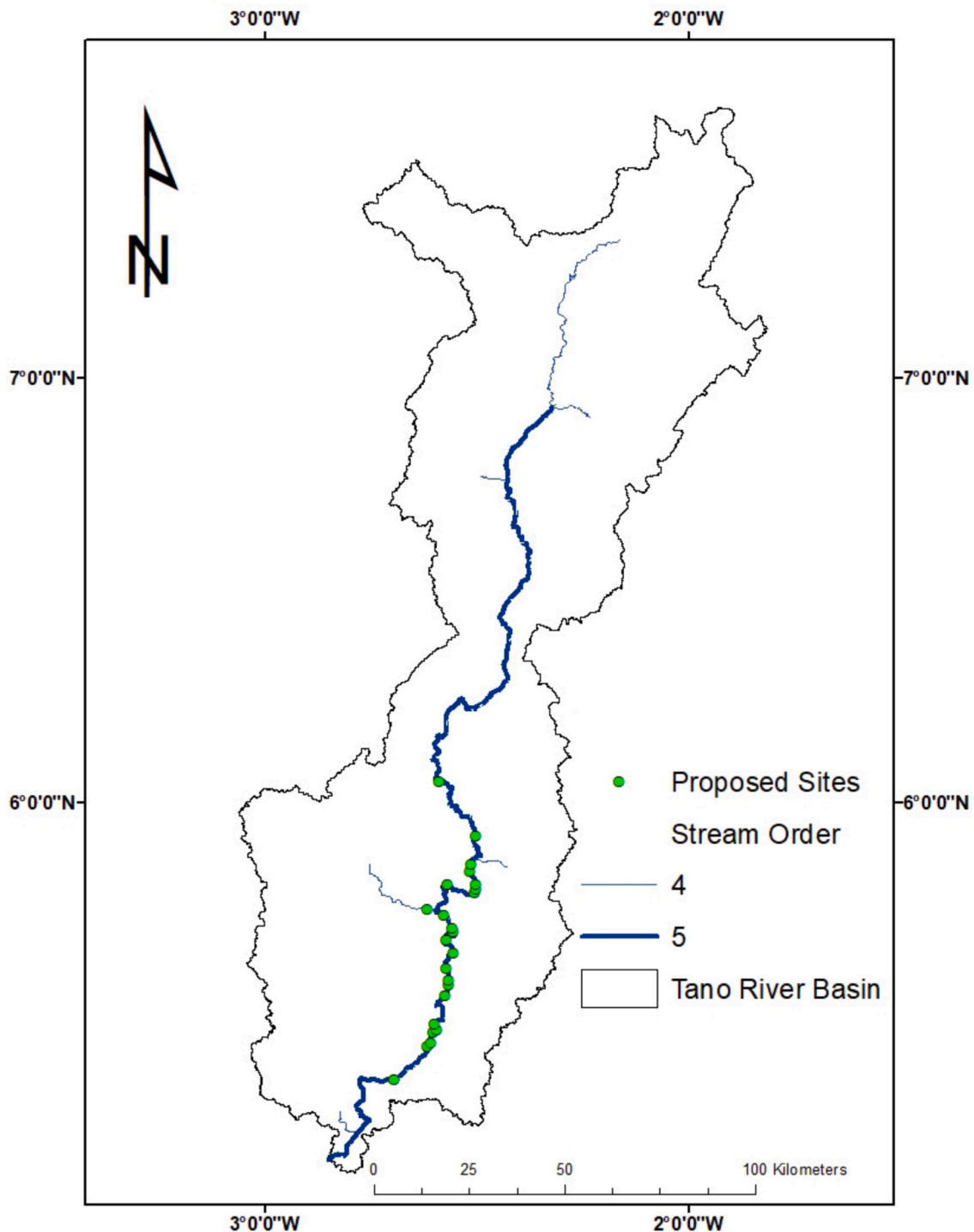


Fig. 9. Proposed sites for power generation.

potential and estimating subsequent energy generation. To enhance analytical clarity, FDCs were developed at monthly intervals, allowing for a more precise characterisation of seasonal flow variations. Table 7 details the estimation of firm power, firm energy and minimum flow requirement at 90 % dependability for each of the 24 candidate hydropower sites. The overall performance of an HP project is influenced by multiple factors, including the efficiency of components such as the turbine, penstock, and generator, as emphasised by Fagbohun (2016).

From the analysis of the hydropower development project from Table 8, 23 out of 24 candidate sites are located on the fifth stream

order, signifying high flows. Only hydropower site fourteen (HP14) is on a lower-order stream (stream order 4). Despite the location of HP14 on the fourth stream order, it is not the least promising site in power and energy production. This is attributable to its relatively high head (16 m). Hydropower site seven (HP7) has the highest firm power at 9.8 MW and firm energy of 46.4 GWh. This site is positioned approximately 7 km upstream from Biaso in the Wasa Amenfi West District.

HP7's high power and energy potential is followed by HP9, HP11, HP3, and HP6 at 9.1 MW and 43 GWh, 9.0 MW and 42.8 GWh, and 8.9 MW and 41.9 GWh respectively. On average, each of the twenty-four

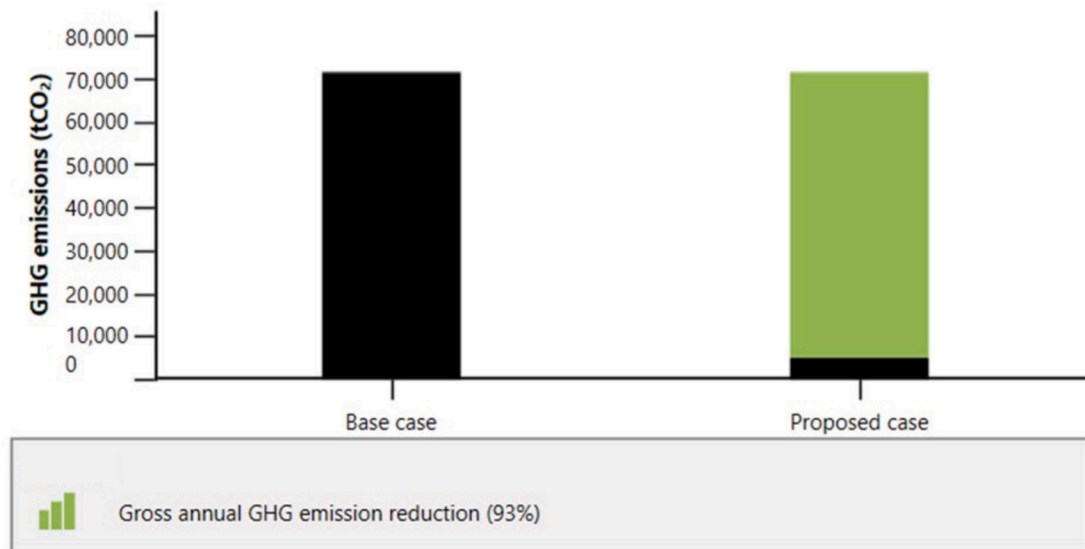
Table 9

Financial summary of the 24 potential hydropower sites.

HP site	Total initial cost (\$)	Gross emissions reduction (tCO ₂ /yr)	Pre-tax IRR equity	Pre-tax IRR asset	Simple payback (year)	Equity payback (year)	NPV (\$)	Discount rate (%)	Project lifetime (year)
1	152,234,000	46,422	17.2 %	6.7 %	9.4	7.6	64,626,904	9	30
2	117,208,000	46,390	27.3 %	10.4 %	7	4.1	102,386,328	9	30
3	133,954,000	63,819	36.1 %	13.1 %	5.9	3	164,448,745	9	30
4	125,454,000	54,409	31.2 %	11.7 %	6.4	3.5	129,933,789	9	30
5	116,661,000	45,054	26.1 %	10 %	7.2	4.3	95,848,509	9	30
6	133,321,000	63,515	36.1 %	13.1 %	5.9	3	163,740,888	9	30
7	168,846,000	57,120	22.3 %	8.7 %	8	5.3	111,635,889	9	30
8	122,729,000	52,696	30.7 %	11.5 %	6.5	3.6	124,515,676	9	30
9	132,095,000	66,619	39.2 %	14.1 %	5.5	2.7	178,496,199	9	30
10	109,479,000	41,213	25 %	9.6 %	7.4	4.6	84,850,199	9	30
11	132,095,000	66,502	39.1 %	14.1 %	5.6	2.7	177,857,919	9	30
12	109,479,000	40,957	24.7 %	9.5 %	7.5	4.6	83,453,035	9	30
13	125,007,000	57,711	34.2 %	12.6 %	6.1	3.2	144,439,053	9	30
14	132,095,000	66,024	38.6 %	13.9 %	5.6	2.8	175,248,329	9	30
15	109,479,000	40,838	24.5 %	9.5 %	7.5	4.7	82,801,427	9	30
16	109,479,000	5327	Negative	Negative	None	None	−111,206,732	9	30
17	109,479,000	36,937	20.1 %	7.9 %	8.5	6.1	61,485,995	9	30
18	125,007,000	51,830	27.9 %	10.6 %	6.9	4	112,311,699	9	30
19	125,007,000	51,733	27.8 %	10.5 %	6.9	4	111,781,925	9	30
20	109,479,000	36,684	19.9 %	7.8 %	8.6	6.2	60,106,215	9	30
21	109,479,000	36,637	19.8 %	7.8 %	8.6	6.2	59,848,338	9	30
22	109,479,000	36,631	19.8 %	7.8 %	8.6	6.2	59,818,328	9	30
23	117,504,000	40,119	19.6 %	7.7 %	8.6	6.3	63,343,692	9	30
24	109,479,000	32,665	15.7 %	6.1 %	9.9	8.7	38,146,317	9	30

GHG emissions

Base case	tCO ₂	71,633	
Proposed case	tCO ₂	5,014	
Gross annual GHG emission reduction	tCO₂	66,619	93%

**Fig. 10.** GHG emission comparison for HP9.

sites has the potential to generate up to 6.6 MW of power and 28.8 GWh of energy in a dry year. Conversely, HP16 exhibits the lowest power potential, with an estimated power output of 0.98 MW and a firm energy of 4.6 GWh. This low power potential is evident from the site's low flow at 90 % dependability. The power and energy values estimated on the 90 % dependability of flows represent the minimum achievable energy output from these sites. The twenty-four potential sites are illustrated in Fig. 9 and detailed in Table 8. This table displays comprehensive details, including each site's coordinates, head, stream order, stream reach, and

slope.

Table 9 provides a detailed financial analysis of the 24 potential HP sites in the Tano Basin, presenting the projected initial costs, gross emissions reductions, IRR for equity and assets, payback periods, NPV, discount rate, and project lifetime. These financial indicators are critical for assessing the techno-economic feasibility of the proposed projects. The total initial costs across all 24 sites range from \$109.4 million (HP 10) to \$172.1 million (HP7). The NPV, indicating the sum of discounted cash flow to the present, also varies, ranging from \$ − 111.8 million

Table 10
Project financial parameters for HP9, HP11 and HP14.

Parameters	Unit	HP9	HP11	HP14
Inflation rate	%	2	2	2
Discount rate	%	9	9	9
Project Life	Years	30	30	30
Finance				
Debt ratio	%	70	70	70
Debt	\$	92,466,500	92,466,500	92,466,500
Equity	\$	39,628,500	39,628,500	39,628,500
Debt interest rate	%	7	7	7
Debt term	Years	15	15	15
Debt payments	\$/Year	10,152,325	10,152,325	10,152,325
Annual Revenue				
Electricity exported to grid	MWh	289,269	288,762	289,269
Electricity export rate	\$/kWh	0.1	0.1	0.1
Electricity export revenue	\$	28,926,912	28,876,183	28,876,183
Electricity export escalation rate	%	2	2	2
Cost Savings Revenue				
Feasibility Study	3.1 %	4,096,000	4,096,000	4,096,000
Development	3.7 %	4,923,000	4,923,000	4,923,000
Engineering	2.1 %	2,748,000	2,748,000	2,748,000
Power System	73.9 %	97,555,000	97,555,000	97,555,000
Balance of system and miscellaneous	17.2 %	22,773,000	22,773,000	22,773,000
Total initial cost	100 %	132,095,000	132,095,000	132,095,000
Annual cost and debt payments				
O&M	\$	5,086,689	5,086,689	5,086,689
Debt payments – 15 yrs	\$	10,152,325	10,152,325	10,152,325
Total annual costs	\$	15,239,014	15,239,014	15,239,014
Annual savings and revenue				
Electricity export revenue	\$	28,926,912	28,876,183	28,668,777
Total annual savings and revenue	\$	28,926,912	28,876,183	28,668,777

(HP16) to \$178.5 million (HP9). While a positive NPV generally signifies strong economic performance, a higher numeric value demonstrates a high-quality investment (Klein & Fox, 2022). Table 9 shows that HP sites like HP20, HP21, HP22 and HP24 show significantly lower NPVs, indicating weaker financial viability. HP 16 is not financially viable owing to its negative NPV.

The IRR is another indicator of economic performance corresponding to the NPV. The RETScreen Expert software computes “Pre-Tax IRR”, representing the accurate interest yield provided by the project's equity or assets over its life before income tax (RetScreen Expert, 2020). Sites with IRR (equity) values exceeding 30 % (e.g., HP3, HP4, HP6, HP8, HP9, HP11, HP13 and HP14) demonstrate strong investment potential, yielding returns well above typical financing thresholds. These sites also show higher NPV values compared to their total initial costs. The simple payback period ranges from 5.5 years (HP9) to 9.9 years (HP24), meaning some sites can recoup their costs quickly, enhancing attractiveness for investors. Sites with shorter payback periods, such as HP3, HP6, HP9, HP11 and HP14, align well with low financial risk. In contrast, HP16 (with no precise payback period) demonstrates high risk and reinforces poor economic feasibility.

The potential HP sites also demonstrate varying potential for emissions reduction, ranging from 5327 tCO₂/yr (HP 16) to 66,619 tCO₂/yr (HP9). High-emission reduction sites align with greater energy production, strengthening their environmental benefits and contributions to climate mitigation targets.

Based on the combined economic indicators of a high NPV, strong

IRR values and shorter payback periods, HP9, HP11 and HP14 emerge as the most viable hydropower sites. For example, HP9 has the highest NPV (\$178.5 million), an impressive IRR (39.2 %) and the shortest payback period (5.5 years) despite it not having the highest theoretical hydropower potential. Notably, HP9 also has the highest gross emissions reduction potential (66,619 tCO₂/yr) making it the most promising HP site to be further developed.

Conversely, HP16, HP22 and HP24 are economically unviable due to negative or low NPVs, high payback periods and low IRRs. Specifically, HP16 exhibits a negative IRR value and lacks a payback period, suggesting an inability to recover investment costs. The low NPVs of HP22 and HP24 show that they are unattractive for investment, which may stem from operational inefficiencies and high initial costs.

Emissions analysis

RETScreen Expert software evaluates the GHG emissions of hydropower projects by comparing a proposed case against a baseline scenario. In this study, the baseline scenario reflects Ghana's energy mix, which is primarily dominated by thermal power generation. For example, in the case of HP9, assuming transmission and distribution losses of 7 % and an emission factor of 0.215 tCO₂/MWh, the estimated annual GHG emissions for the baseline scenario amount to 71,633 metric tons of CO₂. In contrast, the RoR project is expected to reduce emissions by 66,619 metric tons of CO₂ per year (equivalent to avoiding the consumption of 154,927 barrels of oil). This highlights the potential environmental benefits of the proposed HP project.

According to the RETScreen Expert software, GHG emissions from the RoR project arise primarily from the construction, operation, and maintenance phases of the SHP plant. Fig. 10 compares the GHG emission for the baseline and proposed cases for HP9.

Financial viability

The financial assessment of the candidate hydropower sites was conducted with all monetary values denominated in USD to ensure consistency. The analysis assumes a 30-year project lifespan, with a projected annual growth rate of 2 % for the power export rate. The default inflation rate in RETScreen for USD is also set at 2 %. Additionally, operation and maintenance (O&M) costs are expected to increase by 2 % per year to account for future power purchase and sale transactions.

To finance the project, 70 % of the initial capital investment is assumed to be covered through a loan, with the remaining 30 % funded through equity. The loan is structured with a repayment period of 15 years at an interest rate of 7 %. As a result, the loan component for HP9, HP11, and HP14 amounts to \$92,466,500 each, while the equity portion totals \$39,628,500 per site, as detailed in Table 10.

Figs. 11 and 12 visually represent HP9's projected cash flows over the 30-year duration, highlighting the financial feasibility of the proposed hydropower developments.

Conclusion and recommendations

There is a potential for SHP generation within the Tano River Basin. Out of consideration for factors such as head and flows, 24 suitable RoR sites have been identified for the installation of SHP projects in the basin. These locations exhibit a range of maximum and minimum head levels, with maximum heads reaching 16 m and minimum heads at 10 m. To qualify for site selection, specific criteria were applied, including a stream order of four (4) or five (5), at least ten meters in elevation difference (head), along with a separation distance of one thousand meters (1000 m) between each site and intake point.

A SWAT model was developed for the Tano basin, using streamflow data from the Hydrological Service Department of Ghana from 2003 to 2018. The model underwent calibration and validation processes, and

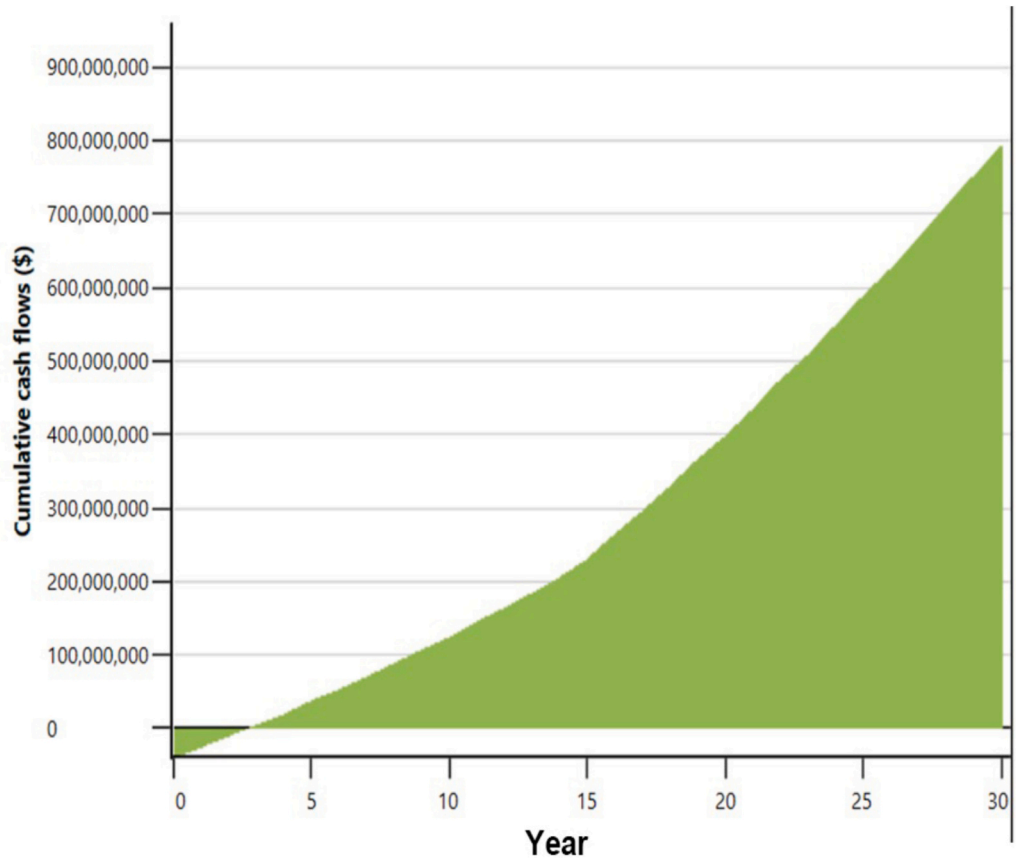


Fig. 11. Cumulative cash flows and duration for equity payback for HP9.

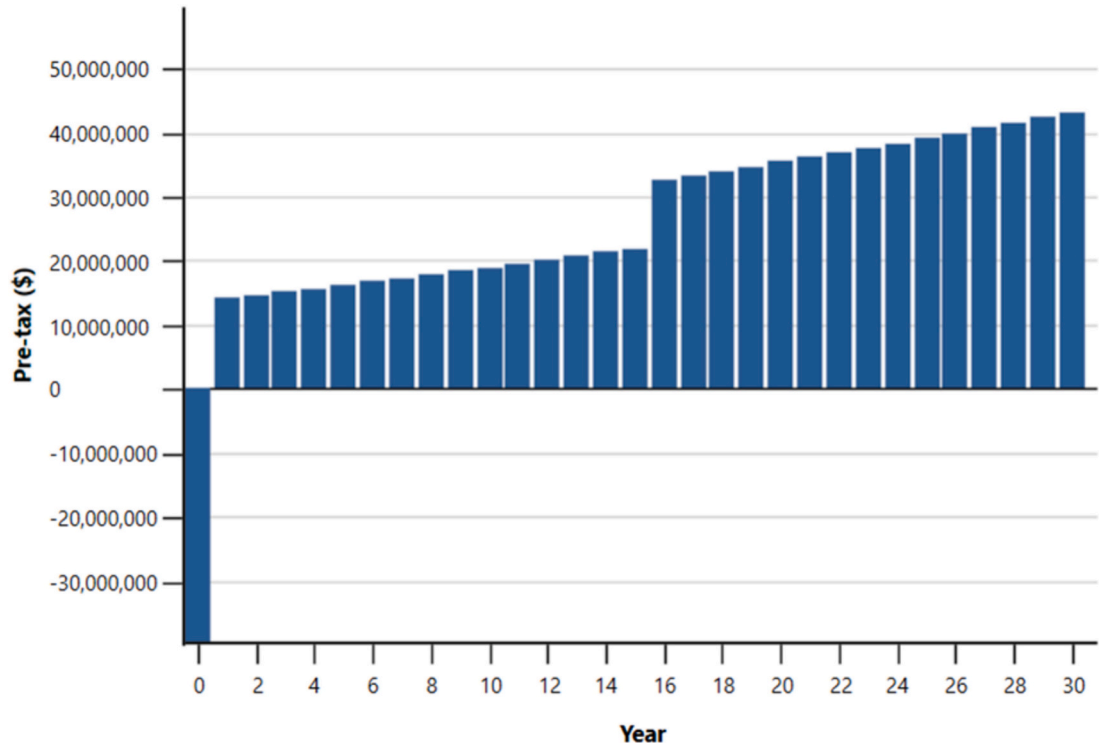


Fig. 12. Yearly cash flows for HP9

the outcomes demonstrated its accuracy. Notably, it yielded NSE values of 0.45 and 0.58, R2 values of 0.47 and 0.60, RSR values of 0.74 and 0.65, and PBIAS values of 8.9 % and 8.2 % for calibration and validation, respectively.

In estimating firm power and energy, FDCs were developed for the proposed sites, focusing on a flow dependability of 90 %. Considering a plant capacity of 60 %, the collective annual firm energy potential from the twenty-four (24) RoR sites varies from 4.2 to 41.8 GWh. Furthermore, the combined firm power output across these twenty-four (24) locations falls from 880.69 to 8841.36 kW. HP7 had the highest theoretical hydropower potential with a firm flow and energy of 9.8 MW and 46.4 GWh. Conversely, HP16 had the least theoretical hydropower potential with a firm power and energy of 0.98 MW and 4.6 GWh.

The RETScreen Expert software was used to assess the economic and emissions-saving potential of all the candidate sites. The findings from the RETScreen assessment revealed that HP9, HP11 and HP14 were the most economically viable projected sites, with NPVs of \$178,496,199, \$177,857,919 and \$175,248,329, respectively. In contrast, HP16 had a negative NPV of \$-111,206,732, demonstrating economic infeasibility. This analysis underscores the need for the government of Ghana to prioritise high-performing sites like HP9, HP11 and HP14 in planning small hydropower projects that address rural electrification challenges. Policymakers and investors should focus on these projects to maximise returns, reduce carbon emissions, and meet ambitious renewable energy targets as this study's findings align with the ECOWAS renewable energy policy of increasing the share of renewable energy by 19 % by 2030 and ultimately the seventh goal of the SDGs.

It is necessary to consider field inspections to enhance the existing findings further and accurately pinpoint potential locations. Relying solely on technical pre-feasibility analyses is insufficient to ensure the success of a SHP project. Additional field inspections will benefit the enhancement of techno-economic assessments and environmental impacts. Future research in this domain should emphasise utilising enhanced data quality, including improved climate data, DEMs, stream networks, land cover and soil maps, and other relevant datasets. Enhanced transboundary cooperation between Ghana and Ivory Coast is crucial for successfully implementing the Tano Basin's proposed SHP development. The Tano River Basin is transboundary. Thus, operational planning and agreements concerning SHP plants in the basin should be fostered to tackle water stress and boost grid reliability between the two nations. It is also advisable to subject the suggested locations to further engineering and geological assessments to ascertain their suitability for SHP development.

Limitations of the study

The primary constraint of this study was the lack of comprehensive data. Streamflow results could have been significantly improved with model calibration at an extra discharge measuring station beyond the Sefwi Wiawso gauging station. Furthermore, the reliance on remotely sensed data due to extensive gaps in the available weather station records posed a constraint. While satellite-derived climate data have demonstrated a certain degree of precision and dependability, they are ideally used to supplement, rather than replace, measured data from weather stations. Additionally, although historical data were utilised to assess HP potential, incorporating future climate scenarios would have yielded more robust results for evaluating the HP potential at the selected sites.

CRediT authorship contribution statement

Godfred Obrempong Addai: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eric Antwi Ofori:** Writing – review & editing, Validation, Supervision, Methodology, Investigation. **Martin Kyereh Domfeh:** Writing – review & editing, Validation, Supervision,

Methodology, Investigation. **Ransford Bakuri Wusah:** Visualization, Validation, Software, Data curation. **Bright Agyemang-Boakye:** Validation, Software, Formal analysis. **Benjamin Ewuley Yankey:** Visualization, Software, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors acknowledge the contributions of the World Bank and Regional Centre for Energy and Environmental Sustainability (RCEES) to this research.

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